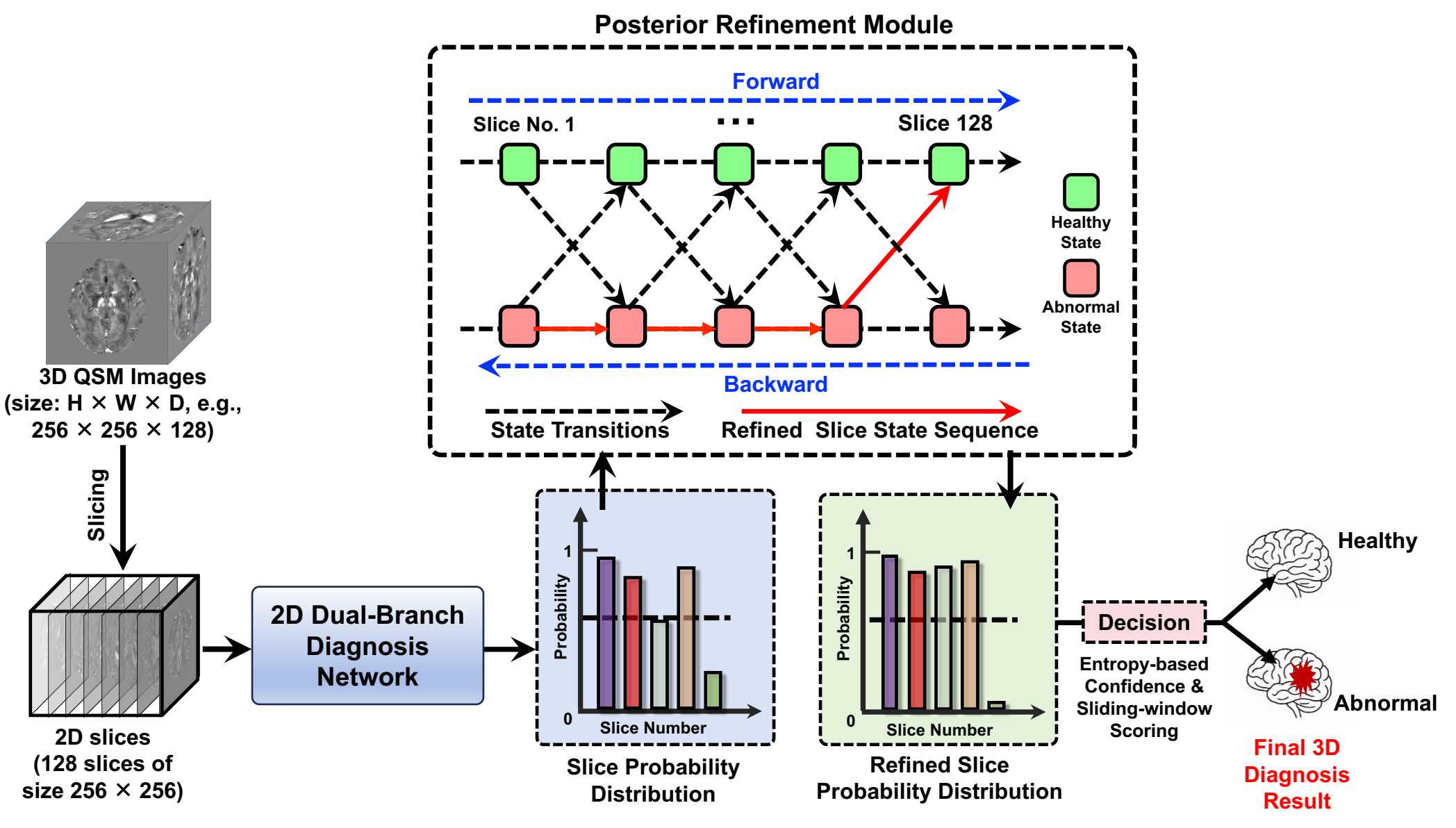

Posterior Refinement Module
Forward
Slice No. 1
Slice 128
Healthy State
Abnormal State
Backward
State Transitions
Refined Slice State Sequence
3D QSM Images
(size: H × W × D, e.g.,
256 × 256 × 128)
Slicing
2D slices
(128 slices of
size 256 × 256)
2D Dual-Branch
Diagnosis
Network
Probability
Slice Number
Slice Probability
Distribution
Refined Slice
Probability Distribution
Decision
Entropy-based
Confidence &
Sliding-window
Scoring
Healthy
Abnormal
Final 3D
Diagnosis
Result